

Georg von Westphalen

NO RULES! 

MEDICAL GENETICS: THE GENETIC CODEBREAKER

SOURCE CODE SABOTAGE & THE BLUEPRINT REBELLION

DIY
OR
DIE

MUTANT
NATION

NO
RULES!

SYSTEM
FAILURE

What the
FLUFF?!

Debugging the blueprint of life. Stay fluffy!

Call
Doc Frizzle!

GENOME
PUNK

MEDELIAN MASTERS

Domi-the-Dominant (Autosomal Dominant)



50% recurrence risk
Vertical transmission
No skipped generations

POWER: The Single-Copy Strike. Domi only needs one corrupted blueprint to take over the entire ring.

LORE: Affects structural proteins/receptors. If Domi is played, the Phenotype-Punch is always visible.

Recessi-the-Hidden (Autosomal Recessive)



25% recurrence risk
Horizontal transmission
Consanguinity risk

POWER: The Stealth-Strike. Hides in the shadows (Carriers) and only attacks when two identical Glitch-Cards meet.

LORE: Affects enzyme factories. One good blueprint keeps the factory functional.

X-the-Carrier-Queen (X-linked Recessive)

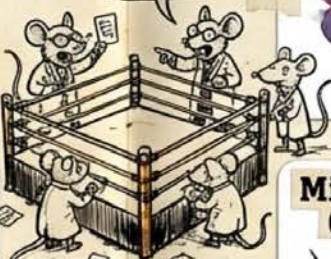


Males affected
Females shielded
No male-to-male transmission

POWER: The Gender-Bias. Protected by her second X-blueprint, but if her sons receive the Glitch-Version, they lose the match.

LORE: The Y-chromosome is a Blank-Page that cannot counter the Glitch.

KayfaBINGO!



Mitochondrial-Molly (Maternal Inheritance)



100% transmission from mother to children
0% from father

POWER: The Battery-Failure. If her code is corrupted, High-Energy-Organs (Brain/Muscle) starve.

LORE: Sperm only give a Nuclear-Blueprint; only the mother's batteries are passed on.



"CHROMOSOMAL CHAOS"



TRISOMY-THE-THREE (Down Syndrome)

STAT:
47, XX/XY, +21
Meiotic non-disjunction
1:700 births

POWER: The Extras-Manual.
Three copies of Manual-21
create a Signal-Overload that
changes the heart and face.
LORE: Causes Endocardial-
Cushion-Cracks. Over-Expression
causes the system-lag.



TURNER-THE-SINGLE-X (Monosomy X)

STAT:
45, X
Female
Short stature
Webbed neck

POWER: The Missing-Page.
Without the second X-blueprint,
the Ovarian-Bench is empty,
and the Aorta-Highway kinks.
LORE: The only survivable human
monosomy. Lack of SHOX-gene
causes the short-stature-stat.



KLINE-THE-EXTRA-X (Klinefelter Syndrome)

STAT: 47, XXY
Male
Tall stature
Infertility

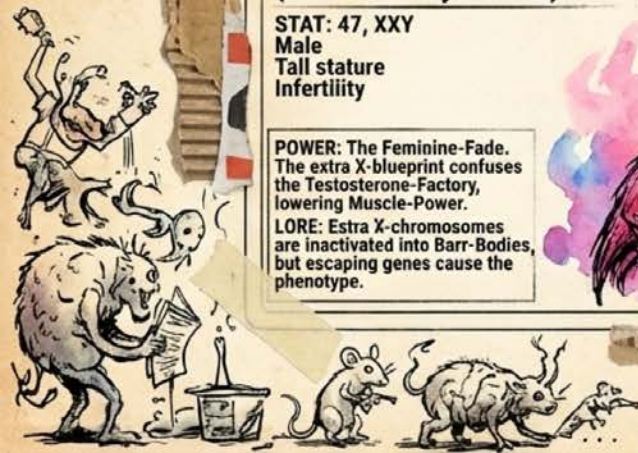
POWER: The Feminine-Fade.
The extra X-blueprint confuses
the Testosterone-Factory,
lowering Muscle-Power.
LORE: Extra X-chromosomes
are inactivated into Barr-Bodies,
but escaping genes cause the
phenotype.



DIGEORGE-THE-DELETER (22q11.2 Deletion)

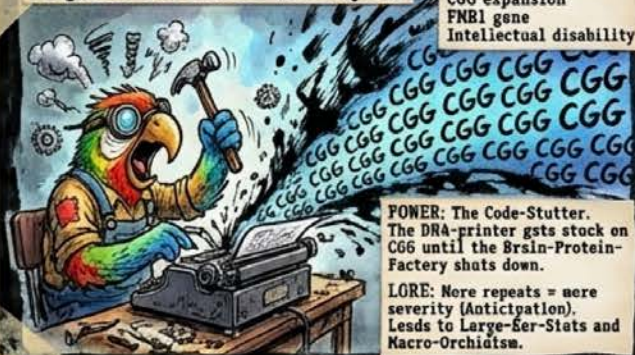
STAT: CATCH-22
(Cardiac, Abnormal face,
Thymic hypoplasia, Cleft
palate, Hypocalcemia)

POWER: The Blueprint-Rip.
A tiny piece of Manual-22 is
torn out, causing multi-system
Neck-and-Chest failure.
LORE: The Immune-Thymus-
Base is missing (3rd/4th
pharyngeal pouches), leaving
the wrestler vulnerable.



REPEAT OFFENDERS

Fragile-X-the-Stutter (CGG Repeat)



STAT:
CGG expansion
FMR1 gene
Intellectual disability

POWER: The Code-Stutter.
The DNA-printer gets stuck on CGG until the Brain-Protein-Factory shuts down.

LORE: More repeats = more severity (Anticipation).
Leads to Large-Ear-Stats and Macro-Orchidism.

Hunting-the-Dance (Huntington Disease)



STAT:
CAG repeats
Chromosome 4
Autosomal Dominant

POWER: The Tesci-Gain.
The CAG-Stutter creates a Rlsfelded-Monster-Protein that kills the Movement-Center.

LORE: Gaiio-of-Function notation. The corrupted protein causes the Chorea-Dance.

Nyo-the-Grip (Myotonic Dystrophy)



STAT:
CTG repeats
DMPK gene
Muscle wasting

POWER: The Mascal-Leck.
The CTG-Stutter prevents muscles from relaxing after a Grip-Rove.

LORE: BHA-toxicity mechanism.
Causes Early-Cataracts and Heart-Block-Glitches.

Fried-the-Ataxia (Friedreich Ataxia)



STAT:
GAA repeats
Frstasin
Autosomal Recessive

POWER: The Iron-Overload.
Sabotages the Ritochondrial-Iron-Manager, causing a Fower-Failurs la nerves and heart.

LORE: The only recessive Repeat-Offsnder. Causes High-Arch-Fee and Cardio-Carl pump failure.

KayfaBOGGLE!



METABOLIC MUTATIONS

PKU-the-Phenyl-Punch (Phenylketonuria)

STAT: Phenylalanine Hydroxylase deficiency / Mousy odor



POWER: The Amino-Acid-Flood. The Breakdown-Factory is broken: Phenyl-Fuel builds up into a Brain-Poison.
LORE: Requires strict Low-Protein-Diet. Beware the Aspartame-Trap.

Marfan-the-Stretcher (FBN1 / Fibrillin-1)

STAT: Autosomal Dominant / Aortic root dilation



POWER: The Elastic-Snap. Weak Tissue-Glue makes the wrestler stretchy but prone to an Aortic Dissection-Burst.
LORE: TGF-beta signaling is over-active because Fibrillin can't Tether it.

CF-the-Clog (Cystic Fibrosis)

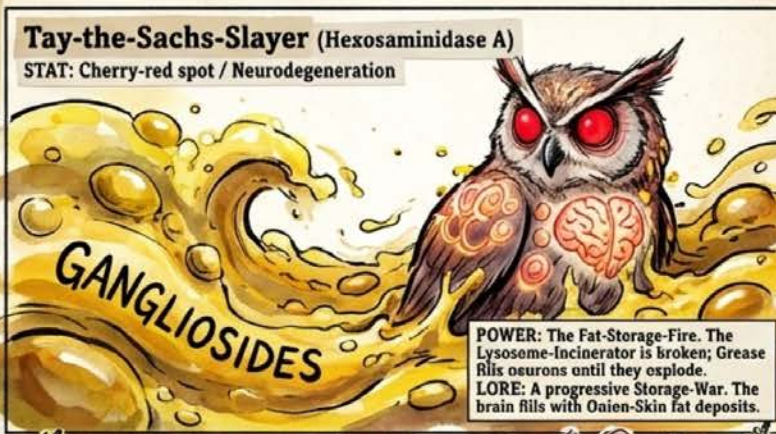
STAT: CFTR gene (delta F508) / Sweat Chloride > 60



POWER: The Salt-Gate-Jam. The Chloride-Tunnel is missing, turning fluids into Sticky-Grey-Glue.
LORE: A protein-folding defect. The tunnel is destroyed before it reaches the surface.

Tay-the-Sachs-Slayer (Hexosaminidase A)

STAT: Cherry-red spot / Neurodegeneration



POWER: The Fat-Storage-Fire. The Lysosome-Incinerator is broken; Grease fills neurons until they explode.
LORE: A progressive Storage-War. The brain fills with Onion-Skin fat deposits.

Diagnosis?!

METABOLIC RIOT!

EPIGENETIC ENIGMAS

Prader-the-Hungry (Prader-Willi)



STAT: Paternal deletion
15q11-13 /
Hyperphagia

POWER: The Eternal-Famine.
The Eatety-Switch is
deleted from the Dad-
Blueprint. Starvation is
an illusion.

LORE: Genomic Imprinting.
Nom-version is normally
silenced; if Dad's is
gone, no one works.

Angel-the-Happy (Angelman)



STAT: Maternal deletion
15q11-13 /
Seizures

POWER: The Puppet-Move.
The Speech-and-Balance-
Script is deleted from
the Mom-Blueprint.

LORS: Reverse of
Prader-Willi. The Dad-
version is naturally
silenced; the Mom-version
is missing.

Methyl-the-Silencer (DNA Methylation)



STAT:
CpC islands /
Gene silencing

POWER: The Code-Lock.
Attaches Lead-Weights to
the DNA-scroll so it
cannot be read.

LORE: Manages Cell-
Specialization. If a
Tumor-Suppressor is
silenced,
Nalignant-Max wins.

Histone-the-Wrapper (Acetylation/Methylation)



STAT:
Chromatin
remodeling

POWER: The
Spool-Tightener.
Decides how tightly
the DNA-Thread is
wrapped to block
Bio-Hackers.

LORE: Acetylation
Loosens the thread
(Dimer-Switch
turns ON).

Diagnosis?!

BLÖDSINN!

CANCER CODE

BRCA-the-Shield-Breaker (BRCA1/2)

STAT: Homologous
recombination
repair



POWER: The Repair-Strike. The DNA-Fiser-Shield is broken; mutations pile up until the arena catches fire.

LORE: Caregiver-Genes. They don't cause cancer directly; they just fall to fix the Double-Strand-Breaks.

P53-the-Judge (Li-Fraumeni)

STAT: Multiple primary
tumors /
Apoptosis
regulator



POWER: The Death-Sentence. If Corrupted, he stops checking for errors and the Mutation-Punks run wild.

LORE: The Guardian of the Genome. Triggers Apoptosis-Arnle if the code is too damaged to fix.

Phil-the-Translocation (BCR-ABL)

STAT:
t(9;22) /
Philadelphia
Chromosome / CML

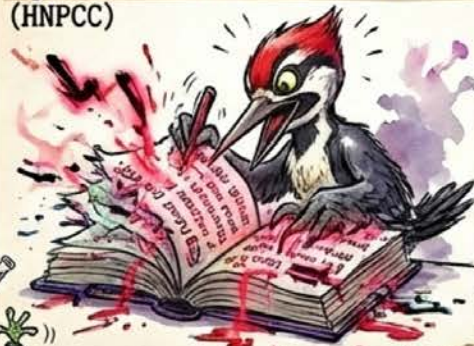


POWER: The Fusion-Weapon. Manual-9 and Manual-22 fuse to create a Monster-KInase-Switch stuck ON.

LORE: Screams at White-Blood-Cells to divide forever. Defeated by Imati-the-Tyrosine-Tamer.

Lynch-the-Mismatch-Glitch (HNPCC)

STAT:
MMR genes /
Ricsatellite
instability



POWER: The Typo-Spreader. The Spell-Checker is broken; every Typo becomes permanent, leading to early Colon-Cancers.

LORE: Not a Polyp-Counter problem, but a base-pair Correction-Failure.

HUMBUG!

Diagnosis?!

BLÖDSINN!

METABOLIC
RIOT!

GW

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INBORN ERRORS OF METABOLISM

Galacto-the-Sugar-Sour (GALT Deficiency)

STAT

Cataracts
Liver failure
E. coli sepsis risk



POWER: The Milk-Poison.
Cannot process Galactose-Sugar; it turns into Toxic-Alcohol that destroys eyes and liver.
LORE: Sugar-Tokens accumulate in the lens, causing Cloud-the-Cataract.

G6PD-the-Fava-Fail (Heinz bodies)

STAT

X-linked
Site cells
Hemolytic anemia



POWER: The Oxidative-Snap.
Without the Protection-Enzyme, oxidative stress causes Red-Cells to explode.
LORE: The Spleen-Sentry takes a Bite out of damaged cells to remove the Heinz Bodies.

Pompe-the-Acid-Storage (Type II Glycogen)

STAT

Acid alpha-glucosidase
Cardiomegaly



POWER: The Muscle-Clog.
Glycogen builds up in Lysosome-Closets until the Heart-and-Muscle-Engines seize up.
LORE: The only Glycogen-Storage-Disease that is also a Lysosomal-Storage-Disease.

Gaucher-the-Bone-Cracker (Glucocerebrosidase)

STAT

Crinkled Paper cells
Erlenmeyer flask feur



POWER: The Marrow-Crowd.
Fat-Filled-Macrophages crowd out the Bone-Marrow-Factory, lowering Platelet-Stats.
LORE: The most common Lysosomal Storage Disease. The macrophages look like Crinkled-Tissue-Paper.



THE GENETIC TOOLKIT

Karyo-the-Map (Karyotype)

STAT
G-banding
Metaphase spread



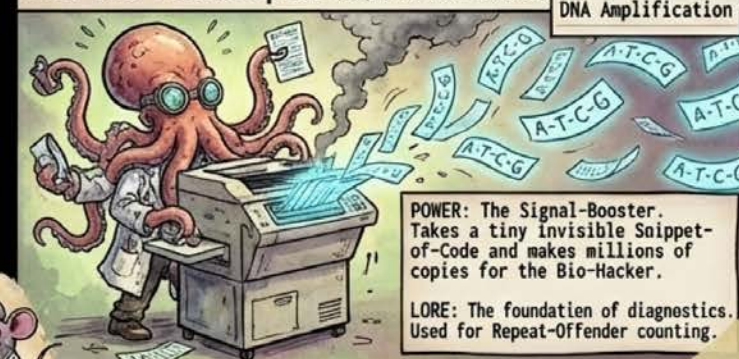
FISH-the-Glowing-Probe (Fluorescence In Situ)

STAT
Microdeletion
detection



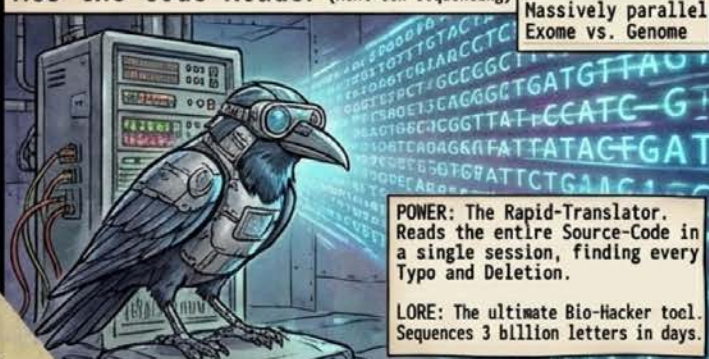
PCR-the-Photocopier (Polymerase Chain Reaction)

STAT
DNA Amplification



NGS-the-Code-Reader (Next-Gen Sequencing)

STAT
Massively parallel
Exome vs. Genome



MedKayfab!



A stylized illustration of a robotic insect character, possibly a wasp or bee, with a metallic body and glowing green eyes. The character is holding a pair of scissors and is cutting a blue DNA double helix. The background is a vibrant, abstract splash of colors including pink, yellow, and green.

Stay fluffy!

T. AAV / Lentivirus

PRENATAL PROTOCOLS

Diagnosis?!

Amnio-the-Fluid-Sample (Amniocentesis)

STAT

15-20 weeks gestation
1% miscarriage risk



POWER: The Early-Blueprint-Look. Draws fluid from the Uterine-Fortress to check the Fetal-Knight's blueprints.
LORE: Samples the shed cells floating in the Amniotic-Sea.

CVS-the-Placenta-Snatch (Chorionic Villus)

STAT

10-13 weeks gestation
First-Look



POWER: The First-Look. Snaps a tiny piece of Placenta-Phil to check the code earlier in the match.
LORE: The placenta and baby share the same Source-Code.

NIPT-the-Blood-Spy (Cell-free fetal DNA)

STAT

Maternal blood screen
cfDNA



POWER: The Stealth-Scanner. Finds tiny Fetal-Code-Scraps floating in the mother's blood.
LORE: No risk to the Fetal-Knight. It's a Screen, not a Diagnostic-Verdict.

AFP-the-Neural-Leak (Alpha-Fetoprotein)

STAT

High in Neural Tube Defects
Low in Down Syndrome



POWER: The Leak-indicator. If the spine isn't closed, AFP Leaks into the mother's blood, triggering the alarm.
LORE: High AFP = Spma-the-Split-Spine. Low AFP = Trisomy-the-Three.

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GENETIC GIANTS & RARITIES

Achondro-the-Dwarf-King (FGFR3 Mutation)

STAT
Autosomal Dominant
80% De Novo



POWER: The Cartilage-Curb. The Growth-Switch is stuck on STOP, preventing bones from stretching.
LORE: A Gain-of-Function in a Negative-Regulator. Short-Limbed but Big-Headed.

Alport-the-Kidney-Coph (Type IV Collagen)

STAT
X-linked
Basket-weave membrane



POWER: The Filter-and-Frequency-Fail. The Basement-Membrane-Mesh is weak; Kidney-Filters leak and Ear-Sensors fail.
LORE: Can't See, Can't Pee, Can't Hear a Bee.

Wilson-the-Copper-King (ATP7B Gene)

STAT
Kayser-Fleischer rings
Low Ceruloplasmin



POWER: The Metal-Poisoning. The Copper-Export-Pump is broken; Copper-Tokens build up in brain and liver.
LORE: Look for the Golden-Rings. Treated by Penicilla-the-Copper-Claw.

Hemo-the-Iron-Man (Hemochromatosis)

STAT
HFE gene (C282Y)
Bronze Diabetes



POWER: The Iron-Rust. The Iron-Gate is stuck open; Rust-Tokens destroy the heart, liver, and pancreas.
LORE: The wrestler turns Bronze and gets diabetes. Treated by Blood-Letting-Leech.

KayfaBIZARRO!

GENETIC PATTERNS

Anticipation-the-Accelerator (Trinucleotide repeats)

STAT: Earlier onset / Increased severity per generation



POWER: The Generational-Speedup. Every time the Repeat-Offender is passed on, the match starts earlier.

LORE: Classic in Hunting-the-Dance. The DNA-polymerase Slips more each time.

Mosaic-the-Two-Face (Mosaicism)

STAT: Multiple cell lines in one individual



POWER: The Split-Persona. Only some of the wrestler's cells have the Trisomy-Glitch, making the phenotype milder.

LORE: Caused by a Post-Zygotic-Error.

Hetero-the-Plasma (Heteroplasmy)

STAT: Mixture of normal/mutated mtDNA



POWER: The Random-Battery. Power-Failure severity depends entirely on the Bad-Battery-Ratio.

LORE: Explains why two siblings with the same mitochondrial disease have totally different Power-Levels.

Imprint-the-Stamp (Genomic Imprinting)

STAT: Parent-of-origin differential expression



POWER: The Parental-Tag. The body Stamps a gene to turn one parental version OFF.

LORE: Explains why Prader-Willi and Angelman look totally different despite being the same Deletion.

What the
FLUFF?!



Call Doc Fizzle!

THE LAB LABYRINTH PREVIEW

G-Banding-the-Stripes (Giemsa Stain)

STAT: Dark (AT-rich) and light (GC-rich) bands



POWER: The Bareods-Reader. Each chromosome has a unique Stripe-Pattern to show niasing chapters.
LORE: Dark bands are Gene-Poor; light bands are Gene-Rich.

Microarray-the-Copy-Counter (CGH)

STAT: Copy Number Variants (CNVs)



POWER: The Quantity-Checker. Compares wrestler's DNA to a Normal-Standard to find Hidden-Deletions or Extra-Duplications.
LORE: Cannot see Balanced-Swaps, but is the king of finding Copy-Huaber-Variants.

Sanger-the-Old-Reader (Chain Termination)

STAT: Gold standard for single variants



POWER: The Letter-by-Letter-Verification. Reads the code one letter at a time to confirm a Single-Typo.
LORE: Slower than RGS, but 100% accurate for confirming a specific Glitch-Spot.

Exome-the-Important-Bits (WES)

STAT: Sequencing only Coding Regions (1%)



POWER: The Efficiency-Hack. Ignores Junk-DNA and only reads the 1% of the code that builds Proteins.
LORE: 85% of disease-causing mutations are found in the Exome. High-Yield-Hacking.



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THE QUESTS & THE ATTENDING (SYNTHESIS)

The Geneticist (Doc FrizzleBob)

STAT:
Board Certified/
Pedigree Master



POWER: The Root-Cause-Finder.
LORE: Treat the code, understand the outcome. Look for the family ghost or the match is a mystery!

Synthesis Quest 1: The Tall Snapper

STAT:
Marfan + Aortic
Dissection



POWER: Clinical Recognition
LORE: Long fingers, high Elastic-Stats + sudden Chest-Bursting-Ache = Identify the Stretcher and the lethal Aortic-Snap.

Synthesis Quest 2: The Two Sides

STAT:
Prader-Willi
vs. Angelman



POWER: The Imprint-Stamp Duel
LORE: Both are Chapter 15 deletions. Explain why the Dad-Deletion vs Mom-Deletion dictates totally different arenas.

Synthesis Quest 3: The Heinz-Bite-Blood

STAT:
G6PD +
Oxidative Stress



POWER: The Enzyme-Fail Matrix
LORE: Fava beans trigger the explosion. Identify the G6PD Fail and the Heinz-Bodies mechanism.

Chromosome 15

Stay fluffy —
and debug
the code!



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